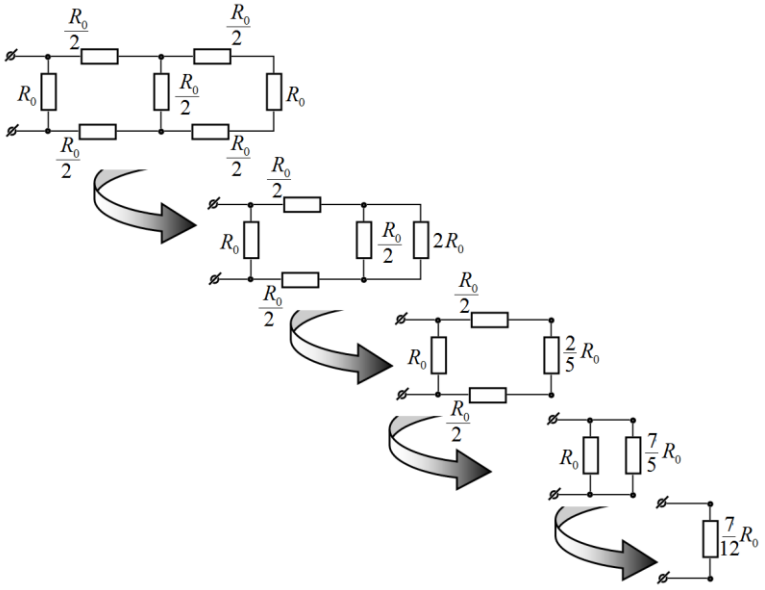


$$R = \frac{7}{12} R_0 . \tag{17}$$

2.2 **[0.2점]**

$$y_k = -x_k , \tag{18}$$

$$x_{N-k} = x_k . \tag{19}$$

2.3 **[1.0점]** .

x_k

$$\frac{x_{k-1} - x_k}{R_0} + \frac{x_{k+1} - x_k}{R_0} + \frac{y_k - x_k}{R_0} = 0 . \tag{20}$$

$y_k = -x_k$

$$x_{k+1} - 4x_k + x_{k-1} = 0 . \tag{21}$$

2.4 **[0.2점]** x_k

. ,

$$x_0 = \varphi_0 , \tag{22}$$

(19) , k .

$k = 0$. N

$$x_N = x_0 . \tag{23}$$

2.5 **[0.2점]** (21) .

(16)

$$x_k = \frac{A q_{N-k} + B q_k}{q_N} = \varphi_0 \frac{q_{N-k} + q_k}{q_N} . \tag{24}$$

2.6 **[0.4점]**

, x_0:

$$I = \frac{x_0 - x_1}{R_0} + \frac{x_0 - x_{N-1}}{R_0} + \frac{x_0 - y_0}{R_0} = \frac{4x_0 - 2x_1}{R_0} . \tag{25}$$

$y_0 = -x_0, \quad x_{N-1} = x_1 .$ $x_0, x_1 ,$